Statistical reasoning — field guide

One dataset → question → design → method → evidence → uncertainty → decision. A formula without its conditions is not an analysis.

1 · DESCRIBE FIRST

Centre: mean uses every value; median resists tails; mode is the most frequent.

Spread: SD is in the data units; variance is squared; IQR is the middle 50%.

Outliers: flag with 1.5×IQR fences; a flag is not proof of error.

Sample vs population: s^2 divides by $n-1$; σ^2 divides by N.

2 · DISTRIBUTIONS

Family	Support	Assumptions	Mean	Variance
Binomial	0…n successes	n fixed, p constant, independent binary trials	np	`np(1-p)`
Poisson	0, 1, 2, … events	λ per fixed interval, independent events	λ	λ
Normal	all real x	μ, σ > 0, continuous symmetric	μ	σ^2

Discrete: masses sum to 1. Continuous: area integrates to 1; $P(X = x) = 0$.

3 · CORE FORMULAS

Sample mean

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

The balancing point of the values. Sensitive to outliers.

Sample variance

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

n−1 corrects the downward bias of squared deviations around the sample mean. Units are squared.

Binomial PMF

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Requires fixed n, independent trials, constant p, and exactly two outcomes per trial.

Normal density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

A density, not a probability. Probability is the area under this curve.

Z-score

$$z = \frac{x - \mu}{\sigma}$$

Signed distance from the mean in standard deviations; σ must be positive.

Poisson PMF

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

λ is the mean count per stated interval; k is a non-negative integer.

Standard error of the mean

$$\mathrm{SE}_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

σ known. With estimated σ, use s/√n and a t critical value when appropriate.

Confidence interval

$$\mathrm{CI} = \mathrm{Estimate} \pm \mathrm{Critical\ Value} \times \mathrm{SE}$$

The interpretation is about the procedure's long-run coverage, not a probability for a fixed parameter.

Test statistic

$$\mathrm{Test\ Statistic} = \frac{\mathrm{Observed} - \mathrm{Hypothesized}}{\mathrm{SE}}$$

The null distribution and sidedness determine how this number becomes a p-value.

p-value

$$p\text{-value} = P(\text{result at least as extreme as observed} \mid H_0 \text{ is true})$$

It is not P(H_0 true), not an overlap of two curves, and it depends on the test and sidedness.

Decision rule

$$p \le \alpha \Rightarrow \text{Reject } H_0, \quad p > \alpha \Rightarrow \text{Fail to Reject } H_0$$

α is fixed before the data. “Fail to reject” is not “accept”.

Simple linear regression

$$Y = \beta_0 + \beta_1 X$$

β_1 is the expected change in Y per one-unit increase in X, in the units of the data.

4 · DECISION RULES

Typical value	Mean (symmetric) or median (skewed / outliers)	Spread	SD with the mean; IQR with the median
Successes in n trials	Binomial	Events in an interval	Poisson
Bell curve / z	Normal	Distribution of sample means	Sampling distribution / CLT
How much a statistic varies	Standard error	Range for a parameter	Confidence interval
Evidence against H_0	Hypothesis test + p-value	Linear association	Pearson r + scatterplot
Causal effect	Randomized experiment; otherwise explicitly limited		

`p ≤ α ⟶ reject H₀ · p > α ⟶ fail to reject H₀ · never “accept H₀”`

5 · TRAPS THAT TRAVEL WITH THE FORMULA

The mean is always the best measure of centre.
Use the median when the distribution is skewed or outliers matter to the question.

Standard deviation and standard error are the same thing.
SD = spread of observations; SE = spread of a statistic across repeated samples.

A p-value is the probability that H_0 is true.
Read it as a compatibility check with the null model, never as the posterior probability of the null.

The p-value is the overlap between the null and alternative distributions.
Shade the equally-or-more-extreme region under H_0; do not shade the alternative.

If $p > \alpha$, we accept H_0.
Say “fail to reject H_0” and describe the uncertainty and the alternative explanations.

You can choose α after seeing the p-value.
Declare α, the test and the sidedness before looking at the data.

Statistically significant means practically important.
Report the effect size and its interval against a pre-specified meaningful threshold.

$r \approx 0$ means X and Y are unrelated.
Look at the scatterplot; r measures linear association only.

A strong correlation proves causation.
Causal claims need design — random assignment, control, or another credible identification strategy.

r=0.8 means Y increases by 0.8 units when X increases by one unit.
Report the slope with units separately from r.

6 · CONTINUE

§1 · The big picture: what is statistics?

§4 · Measures of center

§8 · The binomial distribution

§11 · The central limit theorem

§14 · Confidence intervals

§19 · The p-value

§22 · Type I and Type II errors

§30 · Correlation and Pearson r

§33 · Confounding variables

§40 · The end-to-end checkout case

Source: the owner's local Notion export (43 sections, one HTML file, no figures). This field guide is this site's teaching layer; formulas marked Extension are supplemental. The Poisson λ=20 tail is corrected to 15.68% in §10.